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clear;
clc;

%% TASK 4
% Defining Values
E = 200e9; % youngs modulus of beam material (Pa)
L1 = 0.582; % length of beam A, 1 (m)
L2 = 0.615; % length of beam A, 2 (m)
L3 = 0.645; % length of beam A, 3 (m)
l1 = 0.116; % length of beam B, 1 (m)
l2 = 0.2; % length of beam B, 2 (m)
wA = 0.06376; % width of beam A (m)
wB = 0.06369; % width of beam B (m)
tA = 0.00673; % thickness of beam A (m)
tB = 0.00649; % thickness of beam B (m)
IA = ((wA*(tA^3))/(12)); % moment of inertia of beam A (m^4)
IB = ((wB*(tB^3))/(12)); % moment of inertia of beam B (m^4)

% Defining Flexibility Matrix Elements
a11 = (125*L2^3)/(17496*E*IA);
a12 = (625*L2^3)/(34992*E*IA);
a13 = (925*L2^3)/(34992*E*IA);
a14 = (1255*L2^3)/(34992*E*IA);
a22 = (125*L2^3)/(2187*E*IA);
a23 = (200*L2^3)/(2187*E*IA);
a24 = (275*L2^3)/(2187*E*IA);
a33 = (343*L2^3)/(2187*E*IA);
a34 = (490*L2^3)/(2187*E*IA);
a44 = L2^3/(3*E*IA);
a55 = a44 + ((3*L2^2*L2)/(wA*tA^3*E)) + ((l2^3)/(3*E*IB));
a56 = a44 - ((3*L2^2*L2)/(wA*tA^3*E));

% Mass Matrix (M) and Flexibility Matrix (a)
m = [0.85767, 0.51462, 0.45744, 0.329123, 0.644823, 0.644823]; % mass matrix diagonal elements as row vector (kg)
M = diag(m); % mass matrix
a = [a11, a12, a13, a14, a14, a14;
     a12, a22, a23, a24, a24, a24;
     a13, a23, a33, a34, a34, a34;
     a14, a24, a34, a44, a44, a44;
     a14, a24, a34, a44, a55, a56;
     a14, a24, a34, a44, a56, a55]; % flexibility matrix

% Dynamical Matrix [D]
D = a * M;

% Eigenvalues
[eVectors, eValues] = eig(D);

% Natural Frequencies
lambda = diag(eValues); % eigenvalues in column vector
wn = sqrt(1./lambda); % angular natural frequencies (rad/s)
nf = wn/(2*pi); % natural frequencies (Hz)
[nf, Idx] = sort(nf); % natural frequencies in ascending order (Hz)

% Mode Shapes
mode_shapes = eVectors(:, Idx); % mode shapes, columns (to right) correspond to natural frequency elements (down)

% Display Natural Frequencies and Mode Shapes
disp(num2str(['Natural Frequency (Hz):']));
disp(nf(1:6));
disp(num2str(['Mode Shapes:']));
disp(mode_shapes);

%% TASK 5
% Beam nodes
x_A_nodes = linspace(0, L2*1000, 5); % x-coordinate for beam A nodes (mm) - first isn't a node, it's the base
y_A_nodes = zeros(5,1); % y-coordinate for beam A nodes (mm)
x_B_nodes = ones(3,1) * L2 * 1000; % x-coordinate for beam B nodes (mm)
y_B_nodes = [l2, 0, -l2]*1000; % y-coordinate for beam B nodes (mm) - nodes 5, 4 and 6 respectively

% Beam Coordinates
x = [0, 0, L1, L1, L3, L3, L1, L1]*1000; % all x-coordinates for the beam (anti-clockwise from top left vertex)
y = [34, -34, -34, -200, -200, 200, 200, 34]; % all y-coordinates for the beam (anti-clockwise from top left vertex)
z = zeros(1,8); % all z-coordinates for the beam

% Plotting Mode Shapes for all the natural frequencies
figure;
for i = 1:6
    subplot(3, 2, i);

    % Positive and Negative Mode Shape Projections
    plot3(x_A_nodes', y_A_nodes, [0; mode_shapes(1, i); mode_shapes(2, i); mode_shapes(3, i); mode_shapes(4, i)], 'r', 'LineWidth', 1.2);
    hold on;
    plot3(x_B_nodes', y_B_nodes, [mode_shapes(5, i); mode_shapes(4, i); mode_shapes(6, i)], 'r', 'LineWidth', 1.2);

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plot3(x_A_nodes', y_A_nodes, -[0; mode_shapes(1, i); mode_shapes(2, i); mode_shapes(3, i); mode_shapes(4, i)], 'r--');
plot3(x_B_nodes', y_B_nodes, -[mode_shapes(5, i); mode_shapes(4, i); mode_shapes(6, i)], 'r--');
fill3(x, y, z, 'b', FaceAlpha=0.5)
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% Labels and Titles
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ylabel('mm', 'Interpreter', 'latex', 'FontWeight', 'bold', 'FontSize', 14);
xlabel('mm', 'Interpreter', 'latex', 'FontWeight', 'bold', 'FontSize', 14);
zlabel('Imaginary', 'Interpreter', 'latex', 'FontWeight', 'bold', 'FontSize', 14);
grid on;
grid minor;
title(sprintf('Mode Shape at %.1f Hz', nf(i)), 'FontWeight', 'bold', 'FontSize', 16);
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end
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